

1. What is the magnetic flux through a coil of wire that has an area of $5.0 \times 10^{-2} \text{ m}^2$ and whose plane is perpendicular to a magnetic field of magnitude $2.0 \times 10^{-3} \text{ T}$?
2. A square coil with **85 turns**, measuring **0.15 m** on each side, is in a magnetic field of strength $5.0 \times 10^{-2} \text{ T}$. If the coil is removed from the field in **0.20 s**, what is the average EMF induced in the coil?
3. The back EMF of a motor is **4.2 V** when operated from a **6.0 V** source. When held stationary, the current in the motor's armature is **5.0 A**. What is the resistance of the armature, and what current will exist in the armature when it is rotating at normal speed?

4. Is electromotor more efficient at high frequency or low frequency? Explain.
5. A transformer has **50 primary turns** and **2,000 secondary turns**. The primary coil is connected to a **240 V** source. If the secondary current is **2.5 A**.
- A) What is the secondary voltage?
- B) What is the primary current?
- C) What is the power output of the secondary coil?
- D) What is the power input at the primary coil?